A single exact counterexample to the shifted lonely runner conjecture at $n = 63$, inside the open range $39 \le n \le 94$

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Abstract

The *shifted* lonely runner conjecture (Blanco–Criado–Santos, arXiv:2603.24784, Conjecture 1.1(ii)) says: for distinct positive integer velocities $v_1, \dots, v_n$ and any real shifts s_i there is a time t with $\operatorname{dist}(s_i + v_i t, \mathbb{Z}) \ge \frac{1}{n+1}$ for all i. It is known to fail for $5 \le n \le 17$ and $n \ge 95$ (Poliakova, arXiv:2609.23952), and in our earlier note (doi:10.5281/zenodo.23842167) for $18 \le n \le 38$; the range $39 \le n \le 94$ was open. **We give an explicit exact certificate that the shifted conjecture fails at** $n = 63$: 63 distinct positive integer velocities (largest 128) and rational shifts $a_i/520$ whose 63 open arcs of radius $\frac{1}{64}$ cover the circle. The exact value is $\gamma(v; s) = 66/4225 < \frac{1}{64}$ (gap $\frac{1}{270400}$), so $E_{63} \ge \frac{1}{66}$ in Poliakova's notation, and no more. The certificate is a Poliakova-type move on the base $2, 4, \dots, 128$: six even velocities removed, seven rephased, five odd velocities $1, 3, 9, 27, 81$ added; it is *sporadic*: the same menu is infeasible for $N = n + 1 = 65, \dots, 95$ and $60, \dots, 63$. The certificate passed three exact checkers (two by the producer, one written independently by an auditing AI agent); the covering statement (with injectivity and positivity of the velocities) was kernel-checked in Lean 4 on a pinned toolchain with axioms within the standard whitelist, after a first compile failed on one auxiliary lemma and was repaired (Section 5); no statement lock and no person's check of the statement exist. **This concerns the shifted conjecture at** $n = 63$ **only; nothing is said about the unshifted lonely runner conjecture, and no other** n **in** 39..94 **is settled.**

Markers. [A]: kernel-checked Lean, axioms within {`propext`, `Classical.choice`, `Quot.sound`}. [L]: a cited source, taken as stated. [C]: executed code with an argument written down but not machine-checked. [X]: search output or a recorded absence, supporting nothing by itself. Identifiers `SLR63-·-nnnn` refer to `ledger/ledger.json`.

1 Statement and definition

The shifted conjecture. (BCS [1], Conj. 1.1(ii); definition as pinned in our earlier note, which read it on the sources [1, 3, 2].) For $n \ge 1$, take n runners with *distinct positive integer* velocities $v_1, \dots, v_n$ and shifts $s_1, \dots, s_n \in \mathbb{R}/\mathbb{Z}$, and put

$$\gamma(v; s) = \max_t \min_{1 \le i \le n} \|s_i + v_i t\|, \qquad \|x\| = \operatorname{dist}(x, \mathbb{Z}).$$

The conjecture asserts $\gamma(v;s) \geq \frac{1}{n+1}$ for all such (v,s). A counterexample of size n is therefore a pair with $\gamma(v;s) < \frac{1}{n+1}$. For integer v the function $t \mapsto \min_i \|s_i + v_i t\|$ is 1-periodic and continuous, so its maximum is attained, and

$$\gamma(v;s) < \delta \iff \text{the open arcs } \{t \in \mathbb{R}/\mathbb{Z} : \ \|s_i + v_i t\| < \delta\} \text{ cover } \mathbb{R}/\mathbb{Z}.$$

A single uncovered point (tangency) is *not* a cover. With $\delta = \frac{1}{n+1}$ this is the criterion every checker below uses. Distinctness is essential (with repeated velocities the conjecture fails trivially). In Poliakova's notation [2], $\frac{1}{n+1+E_n} = \inf_{v,s} \gamma(v;s)$, so a certificate gives $E_n \geq 1/\gamma(v;s) - (n+1)$.

Known ranges ([L], `SLR63-L-0001`). Fails for $5 \leq n \leq 17$ [1] and $n \geq 95$ [2]; proved for $n \leq 4$; for $18 \leq n \leq 38$ by our earlier note [4]; Zhang [3] lists the middle range as open. We make no claim beyond $n = 63$.

2 The certificate

`data/certificates/cert_n063.json` (sha256 in `BUNDLE_MANIFEST.txt`): $n = 63$, common denominator $D = 520$, $s_i = a_i/520$, $\delta = \frac{1}{64}$.

- **Fifty-one velocities with shift 0:** the even numbers $2, \ldots, 126$ except $\{4, 8, 12, 16, 24, 32, 36, 48, 64, 72, 96, 108\}$: 2, 6, 10, 14, 18, 20, 22, 26, 28, 30, 34, 38, 40, 42, 44, 46, 50, 52, 54, 56, 58, 60, 62, 66, 68, 70, 74, 76, 78, 80, 82, 84, 86, 88, 90, 92, 94, 98, 100, 102, 104, 106, 110, 112, 114, 116, 118, 120, 122, 124, 126.
- **Twelve velocities with non-zero shift** $a_i/520$, listed in the table (velocities v, numerators a).

v	a	v	a	v	a	v	a
1	385	3	390	9	388	12	264
16	512	24	264	27	391	36	264
48	256	64	8	81	391	128	512

All 63 velocities are distinct positive integers in 1..128 and all $a_i \in [0, 512]$, so $s_i \in [0, 1)$. The shifts are $264/520 = \frac{33}{65} = \frac{1+e}{2}$, $256/520 = \frac{32}{65} = \frac{1-e}{2}$, $512/520 = -e \bmod 1$ and $8/520 = +e$ with the *rephasing offset* $e = \frac{1}{65} = \frac{8}{520}$; the five odd runners sit near $\frac{3}{4}$ ($385, 390, 388, 391, 391$ over 520). Throughout, δ denotes only the cover radius $\frac{1}{64}$; the offset is called e.

3 Exact verification

Three exact checkers (Tier [C]). *Checker 1* (`slr_exact.py`, Python integers; producer): every open-arc endpoint scaled to an integer, greedy sweep of open integer intervals; COVERED. *Checker 2* (`check_cover.c`, `__int128`; producer): the arcs cover iff every arc endpoint lies strictly inside another arc; COVERED, 3961 endpoints checked. These two share an author. *Checker 3* (`audit63.py`, Python standard library plus numpy for a float pre-filter, written from scratch by the auditing AI agent without reading the producer's checkers): exact interval union in integer arithmetic over open intervals with wraparound, and the exact value of γ. A fourth script (`verify63.py`, the main session's own Fractions code, 7858 distinct arc endpoints, none outside every open arc) agrees. We re-ran checker 3 and `verify63.py` for this paper (`data/logs/audit63_run.txt`, `verify63_run.txt`).

Exact value. Among the 331632 pairwise crossing candidates $s_i + v_i t = \pm(s_j + v_j t)$, the maximum of $\min_i \|s_i + v_i t\|$ is, evaluated with exact fractions,

$$\gamma(v;s) = \frac{66}{4225} = \frac{66}{65^2} \approx 0.0156213 \quad \text{at } t^* = \frac{4291}{8450}, \qquad \frac{1}{64} - \gamma = \frac{1}{270400} \approx 3.7 \times 10^{-6}.$$

(The sampled value 0.0156212 in the producer's first report is superseded by this exact one.) The smallest endpoint slack over all arcs is about 3.76×10^{-6}.

Consequence. $1/\gamma = 4225/66 = 64 + \frac{1}{66}$, hence $E_{63} \geq \frac{1}{66}$. **This is all the certificate gives**: it is not $E_{63} \geq 1$ and says nothing about γ_{sh} beyond $\gamma_{\text{sh}}(v) \leq \gamma(v;s) < \frac{1}{64}$.

Negative and tightness checks. The cover *fails* at radius $r = \gamma$ (tangent: an uncovered point exists), holds at $\gamma + 10^{-12}$, and fails at $r = \frac{1}{65}$ (uncovered point at $t \equiv \frac{1}{130}$, so the certificate is not trivially loose). Replacing velocity 81 by 27 (duplicate) is rejected as non-distinct. Perturbing each shift by $\pm\frac{1}{2D}$ breaks the cover in 83 of 126 cases (auditor); the producer's smaller perturbation $\pm\frac{1}{10\sum v}$ broke 3 of 126. With $s = 0$ and velocities $\{1, \ldots, 63\}$ the cover is rejected, as it must be.

4 Structure: a Poliakova-type move

Poliakova [2] starts from $2, 4, \ldots, 2N$ at shift 0 with $N = n + 1$ and radius $\frac{1}{N+1}$; at an odd scale b coprime to 6 she removes $8b, 32b, 192b$, rephases $12b, 24b, 48b, 96b$ to $\frac{1\pm\delta}{2}$ and $16b, 64b$ to $\mp\delta$ and adds $3b, 9b$ at $\frac{1}{4}$; the largest scaled velocity $192b \le 2N$ forces $n \ge 95$. (Her Section 3 and Table 1 differ in the signs for $48, 96, 64$; both variants cover at $n = 95..109$ in our check, `data/code/rule.py`.)

Our certificate has the same shape for $N = 64$ on the base $2, 4, \ldots, 128$ at the *open* radius $\frac{1}{64}$ (which has slack over her closed radius $\frac{1}{65}$):

removed (even, from shift 0)	$4, 8, 32, 72, 96, 108$
rephased (kept, shift changed)	$12, 24, 36 \to \frac{1+e}{2}$; $48 \to \frac{1-e}{2}$; $16, 128 \to -e$; $64 \to +e$
added (odd)	$1, 3, 9, 27, 81$ at about $\frac{3}{4}$

Here the largest rephased velocity is $128 = 2N$ instead of 192, and odd powers of 3 near shift $\frac{3}{4}$ fill the remaining gaps. The count is $64 - 6 = 58$ even runners plus 5 odd ones $= 63 = n$.

How it was found. A CP-SAT set cover (`cpsat_rule.py`): base $2, \ldots, 2N$ at shift 0, radius $\frac{1}{N}$; a pool of 3-smooth multiples of 4 and $\{3, 5\}$-smooth odd velocities may be removed, rephased or added, with shifts $\frac{c}{4} + \frac{j}{8(N+1)}$, $|j| \le 8$, and at most (pool evens)-1 changed runners; exact integer test points (all arc endpoints and cell midpoints inside the part left uncovered by the fixed runners) feed the set cover. Positive control: with Poliakova's rule fixed at $N = 96$ ($n = 95$) the model is feasible and the extracted certificate is COVERED by both producer checkers.

It is sporadic. The *same* pool and menu are infeasible (CP-SAT, exact for that pool and menu) for $N = 95, \ldots, 65$ ($N = 72$ needed 600 s) and for $N = 63, \ldots, 60$; only $N = 64$ is feasible. Why it does not extend uniformly is not understood and no general rule was proved. These infeasibility statements concern only that pool, menu and N; they say nothing about the conjecture, and they were **not re-run by the auditor**. With a wider menu ($J = 16$) infeasibility was found for $N = 47, 46, 44$ and the solver returned UNKNOWN at 120 s for the other tried values; larger pools (5-smooth, all multiples of 4) returned UNKNOWN at 120–400 s (logs in `data/logs/`; Tier [X]). Random local search from an even base found no certificate (best uncovered measure $\approx 10^{-4}$ at several n).

Phase-2 evidence (Tier [X]/[C]; `data/RESULT_PHASE2.md`). A second round looked for further members of the family. Within the pools and shift menu tried (shifts $\frac{c}{4} + \frac{j}{8(N+1)}$, $|j| \le 8$; pools: 3-smooth multiples of 4 with $\{3, 5\}$-smooth odds, 5-smooth, and 3-smooth evens with $\{3, 5, 7\}$-smooth odds), $N = 64$ ($n = 63$) is the only N in 39..95 that admits a cover. The smooth pool with $J = 8$ is infeasible for every other N in 39..95; the 5-smooth pool is infeasible at $N = 47, 46, 45, 43, 39$ and unknown (time limit 120 s) at the remaining values tried; the larger 3-smooth-even pool timed out (even the positive control at $N = 64$). A direct application of the $n = 63$ move with $\delta = 1/(N + k)$, $k = -1..2$, for $N = 64..129$ covers only at $N = 64$, so the move is not uniform in N. This is evidence, not a theorem: unknown cases remain and none of it was re-run by the auditor. The same search also returned a second valid $n = 63$ solution, `data/certificates/cert_n063_alt_sw3_p5.json` (sha256 `dfc8b4d79664a48053aff2b1e3d057235ced913cea483e46b5484bc8fac68d7a`), accepted by both producer checkers (COVERED, 3961 endpoints); it is **not** audited and **not** Lean-checked, and nothing here depends on it. Phase 2 used about 8 core-hours, about 24 in total.

5 Lean

The generated file `data/lean/SLRN63.lean` (imports `Mathlib.Tactic` only, never `import Mathlib`; sha256 in `BUNDLE_MANIFEST.txt`) states, for the typed $v : \mathrm{Fin}\,63 \to \mathbb{N}$ and $s : \mathrm{Fin}\,63 \to \mathbb{Q}$ (the certificate's $a_i/520$):

```
theorem v_injective : Function.Injective v := by decide
theorem v_pos : ∀ i, 0 < v i := by decide
theorem cover : ∀ t : ℝ, ∃ i : Fin 63, ∃ k : ℤ,
    |((s i : ℚ) : ℝ) + ((v i : ℕ) : ℝ) * t - (k : ℝ)| < 1 / 64
```

`cover` is exactly the negation of the conclusion of the conjecture for this (v, s); with injectivity and positivity of v it is a counterexample of size 63. It needs neither γ nor compactness.

What happened, in order. (1) A first full compile of the unmodified file (auditor, pinned Lean 4.34.1 toolchain, 3304 s) returned **rc 1**: `cover` was kernel-accepted (axioms `propext`, `Classical.choice`, `Quot.sound`), `v_pos` accepted (`propext`), but `v_injective` failed (`decide`: maximum recursion depth reached) and so depended on `sorryAx`; the file as a whole did not pass (`data/lean/compile_first_attempt.log`). (2) The fix is `set_option maxRecDepth 1000000 in` before `v_injective`. A small-file test containing v, s and the two small theorems returned rc 0 in 13.6 s with axioms `propext`, `Classical.choice`, `Quot.sound` for `v_injective` and `propext` for `v_pos` (`data/lean/lean_inj.json`). (3) The full recompile of the fixed file (sha256 in `BUNDLE_MANIFEST.txt`) returned **rc 0** in 3327.6 s through `scripts/certified_numerics/lean_pinned.py` (compile plus `#print axioms`; `data/lean/lean_full.json`, `data/lean/compile_full.log`). Axioms: `cover`: `propext`, `Classical.choice`, `Quot.sound`; `v_injective`: the same three; `v_pos`: `propext`. No error is reported, and the source contains no `sorry`, `admit`, `native_decide` or `axiom`. (4) A second, independent run of the same compile (same runner `lean_pinned.py`, same pinned toolchain; the file regenerated from the certificate copy by the same generator and byte-identical to `data/lean/SLRN63.lean`, sha256 `090944f2...`) returned **rc 0** in 2028.2 s with identical axioms (`data/lean/lean_second.json`, `data/lean/compile_second.log`). Both full compiles were run by the main session: the second is a repeat, *not* an independent auditor's compile and *not* a run through `anse/formal/lean_runner.py`. The first attempt's failure and its repair are part of the record, not hidden.

Final gate status. Passed for the compile-and-axioms gate used here, with these limits stated plainly. (a) The compile used the `certified_numerics` runner `lean_pinned.py`, *not* `anse/formal/lean_runner.py`. (b) The statement is *not* locked by LeanMaster's statement-lock tool. (c) The Lean file was produced by our untrusted generator; this does not weaken the kernel result, because the kernel re-checks every case of `cover`, and distinctness (`v_injective`) and positivity (`v_pos`) of v are theorems in the same file. (d) A kernel-checked statement certifies that statement, not its gloss: the claim is the stated covering of the circle by the 63 open arcs, nothing beyond it.

The Lean covering statement is filed at Tier [A] as `SLR63-A-0001` (with an `ai_audit` field; the gate field `audit` stays null). As in the earlier note, no person has certified that the Lean statement means what its gloss says, there is no statement lock, and the repository's `lean_runner.py` gate is separate from the compile used here.

6 Related work and novelty

- Blanco–Criado–Santos [1] (arXiv:2603.24784): the shifted conjecture and failures for $n = 5..17$.
- Poliakova [2] (arXiv:2609.23952): failure for $n \geq 95$ and the move above; she states $18 \leq n \leq 94$ open.
- Zhang [3] (arXiv:2608.13599): lists the middle range as open.
- Our earlier note [4] (doi:10.5281/zenodo.23842167): certificates for $18 \leq n \leq 38$ by a different (local search) method.

Searches on alphaXiv (recency windows after 2026-03-01 and after 2026-09-20, producer, 2026-11-07; two queries including one after 2026-10-01, auditor, 2026-11-08) and a web search found no paper covering any n in 39..94 and none proving the conjecture there. **To our knowledge** this is the first published certificate in that range; this is not a priority claim. Residual risk: arXiv listings after the alphaXiv index, unpublished computations, and private communications. We have not contacted the cited authors. We reproduce no third-party text.

7 Audit corrections applied

The independent audit (`data/AUDIT.md`, §6) required four corrections to the producer's `RESULT.md`; all are applied in the copy in `data/` and in this paper:

1. Shift of velocity 48 is $256/520 = \frac{1-e}{2}$, not $264/520$; the rephasing offset is named $e = \frac{1}{65}$ so that δ means only the cover radius $\frac{1}{64}$.
2. The sampled $\max_t \min_i \| \cdot \| = 0.0156212$ is replaced by the exact $\gamma = 66/4225$ and gap $1/270400$.
3. Added: the certificate gives $E_{63} \geq \frac{1}{66}$ only.
4. The Lean wording: the first compile failed on `v_injective` (recursion depth), with no gate pass for that file; the repaired file passed the full recompile (Section 5).

8 Claim ledger

`ledger/build_ledger.py` files the claims in `ledger/ledger.json` with content-addressed evidence in `ledger/evidence/`: [L] (definition and prior results), [C] (three exact checkers; controls; the headline `SLR63-C-0003`, filed C because an assembly argument is capped at [C]), [X] (search absences; scoop check; what was not done), and one [A] row, `SLR63-A-0001`: the Lean covering statement `cover`, with `v_injective` and `v_pos`, kernel-checked with axioms within the whitelist (not its physical or conjecture-level meaning beyond the stated covering). **The finding itself stays Tier [C]** (assembly argument, capped at [C]). The [A] row is flagged by the gate as unaudited by a person. The Elenchus gate (`ledger/gate_output.txt`) reports 8 claims, no blocking finding and one flag, `LEDGER_UNAUDITED_TIER_A` on `SLR63-A-0001`, exit code `1`: the honest state, since no person has certified the statement.

9 Limitations

- **Shifted conjecture only.** Nothing here concerns the unshifted lonely runner conjecture. One value of n (63); no other n in 39..94 is settled, and no uniform rule was found.
- $E_{63} \geq \frac{1}{66}$ **only**, an upper bound $\Gamma(63) < \frac{1}{64}$; no exact γ_{sh}.
- **Sporadic.** Infeasible for the same pool at $N = 65..95$ and $60..63$; why is not understood.
- **AI producer and AI auditor; no human audit**, no statement lock.
- **Negative sweeps unverified by the auditor**: the CP-SAT infeasibility runs were not re-run, nor `cpsat_rule.py 64 8 60`. The Poliakova rule was re-derived from the paper text by the producer and auditor only.
- **Literature search** covers alphaXiv and one web search; novelty is "to our knowledge" only. Premises about prior ranges are taken as stated ([L]).
- **Lean:** the first full compile failed on `v_injective` (fixed by `set_option maxRecDepth 1000000`); the full recompile passed (rc 0, axioms whitelisted) and a second run of the same compile (same runner and file, main session, 2028.2 s) agreed; it was not compiled by an independent auditor. Not run through `anse/formal/lean_runner.py`; no LeanMaster statement lock; generated by an untrusted generator (the kernel re-checks every case); no person has certified that the statement means what its gloss says.
- **Phase-2 evidence** is a set of exact infeasibility results for specific pools and a shift menu plus UNKNOWN cases; it supports no claim beyond them. The alternative $n = 63$ solution is unaudited and not Lean-checked.
- **Compute.** About 15–16 core-hours in phase 1 and about 8 in phase 2 (about 24 in all; upper bound; CP-SAT counted as wall×6 or 7), cores 1–7; a host reboot occurred during the run and the final sweep completed.

10 Reproducibility

Repository AutoevolveAI, branch `worktree-openai-math-discovery`; HEAD when this paper was written: commit `3fa606c`; the paper and data are uncommitted working-tree files at that time. Preregistration: `data/PREREG_ITEM.md` (hash in `PREREG_ITEM.md.sha256`), written before any search. Copies of all scripts are in `data/`.

```
python3 -I data/code_auditor/audit63.py data/certificates/cert_n063.json   # exact cover + exact gamma
python3 -I data/code_main/verify63.py            # edit the certificate path inside the file
python3 -I data/code/slr_exact.py data/certificates/cert_n063.json   # producer checker 1
python3 -I ledger/gen_table.py ; python3 -I ledger/build_ledger.py
```

The C checker is stored as `data/code/check_cover.c.txt` (rename to `.c` to compile).

References

[1] Blanco, Criado, Santos, arXiv:2603.24784 (v4): the shifted lonely runner conjecture.

[2] Poliakova, arXiv:2609.23952: counterexamples to the shifted lonely runner conjecture for $n \geq 95$.

[3] Zhang, arXiv:2608.13599.

[4] X. Callens, *Exact counterexample certificates to the shifted lonely runner conjecture for* $18 \leq n \leq 38$, doi:10.5281/zenodo.23842167.